

Mohammad Adab

Phone: +91-6006711641 | Email: adabismail000@gmail.com | Portfolio | LinkedIn: adab-ismail | GitHub: adabismail

Experience

Software Development & Research Assistant

Feb 2026 – Present

GAASH Lab, NIT Srinagar

- Eliminated manual record-keeping for **500+ PhD scholars** on schedule by delivering a production Django + PostgreSQL ERP with role-based access control within a cross-functional team.
- Automated B.Tech admissions intake to **zero manual entry errors** across departments by building Django REST Framework APIs within a sprint-based, collaborative Git workflow.
- Shipped **2 production-ready PyTorch vision models** within lab timelines by implementing and benchmarking deep learning pipelines that directly informed active research product decisions.

Frontend Engineer Intern

May 2026 – Present

Traxevo Web Limited (Remote, Delhi)

- Reduced page load time by **40%** for **15,000+** monthly users by engineering high-performance React.js interfaces with optimized Hooks, lazy loading, and REST API integration.
- Increased development efficiency by **30%** by designing a reusable component library and streamlining Agile sprint cycles while maintaining cross-browser compatibility.

Computer Vision & Deep Learning Intern

Dec 2025 – Mar 2026

IIT Roorkee

- Surpassed YOLOv5 baseline by **+5.8% mAP** and **+4.2% F1-score** (mAP@0.5 = **0.813**, precision **88.4%**) by engineering a hybrid ResNet-50 + Vision Transformer detection architecture.
- Cut false positive rate by **18%** against baseline by iterating on backbone depth and attention configurations through structured PyTorch and Scikit-learn experiments.

Projects

ReviewFlow AI — Agentic Customer Review Intelligence

GitHub

Python · LangChain · LangGraph · FLAN-T5 · Azure AI · LCEL

- 80% reduction** in manual review triage at **<15s** response times by building a stateful **4-node** LangGraph agentic workflow with structured LCEL prompt chains.
- 90% sentiment accuracy** on held-out data by engineering FLAN-T5 prompt templates with chain-of-thought reasoning and structured output parsing.
- Negative reviews escalated to stakeholders in **<2 seconds** by wiring Gmail SMTP HTML alerts directly into the agentic decision pipeline.

FlightPulse — Real-Time Flight Deal Monitoring Platform

GitHub

React 18 · Django DRF · PostgreSQL · SQL Server · Celery · Redis · Docker · Nginx

- Sub-**200ms** API response times across **20+** endpoints under concurrent load by architecting a full-stack React 18 + Django system with optimized relational SQL queries.
- 5,000+** daily flight tasks processed without data loss over a **30-day** window by engineering an async Celery + Redis pipeline on real-time Amadeus API data.
- Zero-downtime deployments across all release cycles by containerizing **5** Docker microservices behind an Nginx reverse proxy with health-check orchestration.

Hydra — Distributed MapReduce Engine

Github

Python · FastAPI · REST · Multithreading · React · Distributed Systems

- Built a **fault-tolerant distributed MapReduce** engine on a **Master/Worker** model, scheduling parallel map and reduce tasks across independent **OS-process workers** over a **REST/HTTP** control plane with thread-safe **concurrent** coordination.
- Implemented **heartbeat-based failure detection** with automatic **task reassignment**, lost-output recomputation, and **idempotent stale-result rejection**, guaranteeing correct **distributed key-value aggregation** with **zero data loss** under mid-job worker failure.
- Engineered a **file-based shuffle** with deterministic **CRC32 hash partitioning** and a real-time **React dashboard** monitoring **job lifecycle, worker health**, and live cluster events.

Technical Skills

Languages	Python · Java · JavaScript · C · SQL
Frontend	HTML · CSS · React.js · Bootstrap · Tailwind CSS
Backend	Django · Flask · FastAPI · REST APIs · Celery
Databases	PostgreSQL · SQL Server · MongoDB · Redis · ChromaDB · FAISS
DevOps & Tools	Git · Docker · Nginx
AI / ML	PyTorch · LangChain · LangGraph
Core CS	Data Structures & Algorithms · OOP · Operating Systems · DBMS · Computer Networks

Education

B.Tech in Information Technology

Aug 2023 – Present

National Institute of Technology Srinagar

CGPA: **7.99/10**